MESSMAP

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

The requirements below describe observable system behavior. IDs such as FR-01 can be used by the team when discussing implementation and testing.

1. Authentication and Role Management

ID	Requirement	Expected behavior
FR-01	Student registration	The system shall let a student create an account using the required registration fields.
FR-02	Owner registration	The system shall let a mess owner create an owner account.
FR-03	Login/logout	The system shall authenticate registered users and provide a logout function.
FR-04	Role-based access	The system shall direct users to the correct dashboard and prevent access to unauthorized role features.
FR-05	Input validation	The system shall validate required fields and show understandable error messages.

2. Student Features

ID	Requirement	Expected behavior
FR-06	Discover nearby messes	The system shall display approved mess listings near the student's current or selected location.
FR-07	Location permission	If location access is denied or unavailable, the system shall explain the issue and provide a manual-location alternative where implemented.

FR-08	Search	The student shall be able to search messes by name or supported search criteria.
FR-09	Filter and sort	The student shall be able to filter results by supported criteria such as distance, price, food type, meal availability, and rating.
FR-10	View mess details	The system shall display the mess profile, menu, price, food type, operating hours, availability, contact details, location, and reviews when available.
FR-11	View map/directions	The system shall show the mess location on a map and provide a route/directions link when supported by the map provider.
FR-12	Compare options	The student shall be able to compare available information across messes; a dedicated comparison screen is optional unless selected for the MVP.
FR-13	Submit review/rating	The student shall be able to submit a rating and review subject to the project's review rules.
FR-14	View reviews	The student shall be able to read reviews and ratings associated with a mess.

3. Mess Owner Features

ID	Requirement	Expected behavior
FR-15	Submit mess listing	The owner shall enter mess name, address/location, contact details, food type, pricing, timings, and other required details.
FR-16	View verification status	The owner shall be able to see whether a listing is Pending, Approved, or Rejected.

FR-17	Edit profile	An owner shall be able to edit only the mess listing linked to their account.
FR-18	Manage menu	The owner shall be able to create or update daily/weekly menu information.
FR-19	Manage prices	The owner shall be able to update daily meal prices and/or monthly plan prices if those fields are included.
FR-20	Manage availability	The owner shall be able to update meal availability, operating status, holidays, and temporary closure information.
FR-21	Manage timings/contact	The owner shall be able to update opening hours and contact details.
FR-22	Resubmit rejected listing	The owner shall be able to correct rejected information and submit it again for verification.

4. Administrator Features

ID	Requirement	Expected behavior
FR-23	Admin login	The system shall provide a protected admin login.
FR-24	Manage users	The administrator shall be able to view users and take permitted actions such as activating or blocking an account.
FR-25	Manage owners	The administrator shall be able to view and manage mess-owner accounts.
FR-26	Verify listings	The administrator shall review submitted listings and approve or reject them, optionally recording a reason.

FR-27	Manage listings	The administrator shall be able to inspect, deactivate, or remove invalid listings.
FR-28	Moderate reviews	The administrator shall be able to inspect reported or inappropriate reviews and take moderation action.
FR-29	Dashboard overview	The administrator shall be able to view useful counts or lists of users, owners, pending listings, and reviews.

5. Notifications and Error Handling

ID	Requirement	Expected behavior
FR-30	Status feedback	The system shall confirm successful actions and explain failed actions.
FR-31	Registration/verification notification	If email notifications are included, the system shall notify users of relevant registration or listing-status events. Email integration must be confirmed as part of scope.
FR-32	Empty results	The system shall display a helpful message when no messes match the search or filters.
FR-33	Network/server error	The system shall show a user-friendly error and avoid exposing internal stack traces.

6. Key End-to-End Use Cases

Use Case UC-01: Student Finds a Mess

Student opens MessMap and logs in if login is required for the chosen action.

Student allows location access or selects a location manually.

System retrieves approved, active mess listings matching the location and selected filters.

Student opens a mess profile and reviews its menu, pricing, timings, availability, location, and reviews.

Student contacts the mess or opens directions.

Use Case UC-02: Owner Registers a Mess

Owner registers/logs in and opens the owner dashboard.

Owner enters mess details and submits the listing.

System validates the information and stores the listing with Pending status.

Administrator reviews the listing and approves or rejects it.

System updates the status. An approved listing becomes visible to students; a rejected listing can be corrected and resubmitted.

Use Case UC-03: Admin Moderates the Platform

Administrator logs in and opens the admin dashboard.

Administrator reviews pending mess listings, user/owner accounts, or reported reviews.

Administrator approves/rejects listings or performs an authorized management action.

System records the change and updates the relevant status or listing visibility.